Provenance of 'Einstein-Rosen bridges in the gravitational fields of dirac16complex'

1. Sources

- A. Einstein and N. Rosen, "The Particle Problem in the General Theory of Relativity", Phys. Rev. 48, 73-77 (1935), doi:10.1103/PhysRev.48.73. Quoted in Pre-gravity6plane.nb (cells 1397-1399, 1406, 1412): the substitution $u^2 = r - 2m$, two congruent sheets $u > 0$, $u < 0$, joined by the hyperplane $u = 0$ in which g vanishes, called a bridge. The charged bridge with reversed sign of the electromagnetic energy and $m = 0$ is from the same paper; here it is re-derived as $r^2 = u^2 + q^2$, $ds^2 = -\frac{u^2}{u^2+q^2}dt^2 + du^2 + (u^2+q^2)d\Omega^2$ and verified.
- Pre-gravity6plane.nb (P. L. Nash, (C) 2022, GPL): primordial vielbein $\mathrm{diag}(\tan 6Hx_0,\ e^{-a_4}\sin^{-1/6}6Hx_0\ (\times 3),\ 1,\ e^{a_4}\sin^{-1/6}6Hx_0\ (\times 3))$ (cells 901, 913) and the domain $0 < 6Hx_0 < \pi/2$ (cell 48).
- The modern and future fields are this work's definitions ($f = 1 - r_s/x_0$, $f = 1 - r_s/x_0 - H_\Lambda^2 x_0^2$ for the (x_0, x_4) metric $dx_0^2/f - f\,dx_4^2$).
- Standard results used in Section 4: W. G. Unruh, Phys. Rev. D 14, 870 (1976); W. Israel, Phys. Lett. A 57, 107 (1976); J. B. Hartle and S. W. Hawking, Phys. Rev. D 13, 2188 (1976); Kruskal coordinates (M. D. Kruskal, Phys. Rev. 119, 1743 (1960)). The picture of bridges as pair correlations follows J. Maldacena and L. Susskind, Fortsch. Phys. 61, 781 (2013).

2. Computations (Wolfram Language 15.0.1, exact, 2026-10-30)

result	6planes.nb
neutral bridge: $R_{\mu\nu} = 0$, $\det g = -4u^2(u^2+2m)^4\sin^2\theta$, even in u, Kretschmann $48m^2/(u^2+2m)^6$	Part 8, Out[31]
charged bridge: $G_{\mu\nu} + 8\pi T^{EM}_{\mu\nu} = 0$, $\det g(0) = 0$, Kretschmann$(0) = 56/q^4$	Part 8, Out[31]
modern and future bridges: $g_{uu}(0)$, $g_{44}(0)$, evenness, $R(0)$, κ (exact cubic roots; check at $u = 10^{-6}$)	Part 8.1, Out[32]
primordial: $R = -\frac{3H^2}{2}(57 + 64\cos 12Hx_0 + 3\cos 24Hx_0)\csc^4 6Hx_0$, series $-\frac{31}{216H^2x_0^4}$, $\det g = \sec^2 6Hx_0$	Part 8.2, Out[33]
continuation factor $e^{-\pi\epsilon/\kappa}$, occupation $1/(e^{2\pi\epsilon/\kappa}+1)$ equals Fermi-Dirac at $T_H = \kappa/2\pi$	Part 8.3, Out[34]

The curvature routines (Christoffel symbols, Riemann tensor $R^i{}_{jkl} = \partial_k\Gamma^i{}_{jl} - \partial_l\Gamma^i{}_{jk} + \Gamma^i{}_{km}\Gamma^m{}_{jl} - \Gamma^i{}_{lm}\Gamma^m{}_{jk}$, Ricci tensor $R_{jl} = R^i{}_{jil}$, Einstein tensor, Kretschmann scalar) are written out in 6planes.nb and were first tested on the Einstein-Rosen metrics in a scratch script (test_er.wls) with identical results.

3. New in this work

Application of the Einstein-Rosen substitution $x_0 = x_h + su^2$ to the modern and future fields; the proof that the primordial field's degenerate hypersurface is singular; the identification of the Kruskal pairing with the $\{+\epsilon, -\epsilon\}$ universes of dirac16complex.